

Strategic Objectives & Key Results

Safety Excellence

Zero gas-related public incidents; maintain pipeline integrity index >98%; sub-30 min emergency response

Operational Excellence

Network availability $\geq 99.9\%$; reduce unplanned outages by 40%; gas loss <1% of throughput

Energy Transition

Integrate 20% hydrogen-blend-ready networks by 2030; expand biomethane injection capacity

Infrastructure Modernization

Replace aging cast iron/steel mains with PE pipe; deploy smart pipeline monitoring

Customer Centricity

CSAT >85%; digital engagement >70%; 24/7 gas emergency response capability

Digital Transformation

Smart metering 100% deployment; AI-driven pipeline integrity & leak detection

Financial Performance

RAB growth 5-7% p.a.; operating efficiency ratio <72%; WACC-aligned returns

Environmental Stewardship

Reduce methane emissions 50% by 2030; net-zero operations by 2040

Business Capability Map

Customer Operations

- Customer Enrollment & Onboarding
- Billing & Payment Processing
- Customer Service & Support
- Complaint & Dispute Management
- Energy Efficiency Programs
- Gas Safety Awareness Programs

Gas Supply & Procurement

- Natural Gas Procurement & Contracting
- LNG / LPG Procurement & Storage
- Gas Supply Portfolio Management
- Biomethane & Green Gas Sourcing
- Supply Balancing & Nominations
- Gas Quality Management

Transmission & High-Pressure Operations

- Transmission Pipeline Operations
- Compressor Station Operations
- Gas Receiving & Metering Stations
- Pressure Regulation (Gate Stations)
- Gas Flow Scheduling & Control
- Interconnection & Balancing

Distribution Network Operations

- Low & Medium Pressure Network Management
- District Pressure Regulation
- Service Connection Management
- Odorization Management
- Network Monitoring & SCADA
- Emergency Response & Gas Escapes

Asset & Pipeline Integrity

- Pipeline Integrity Management (PIMS)
- Corrosion Protection & Cathodic Management
- Inline Inspection & Pigging
- Asset Lifecycle Planning
- Capital Works & Mains Replacement
- Third-Party Damage Prevention (Dial Before You Dig)

Safety & Emergency Management

- Gas Escape & Emergency Response
- Hazardous Area Classification
- Major Accident Hazard (MAH) Management
- Safety Case & Risk Assessment
- Workforce Safety & Permit to Work
- Business Continuity Planning

Regulatory & Compliance

- Gas Safety Regulatory Compliance
- Pipeline Code Compliance (ASME B31.8)
- Environmental License Management
- Tariff & Rate Case Management

Regulatory Reporting & Audits

Technology & Data

IT/OT Infrastructure Management

Application Portfolio Management

Cybersecurity & OT/ICS Security

Data Management & Analytics

Smart Metering & IoT Platform

Digital Twin & Simulation

Core Value Streams

1. Connect Customer to Network

Application



Feasibility



Design



Construction



Purge & Test



Commissioning



Activation

2. Transport & Deliver Gas

Supply Nomination



Receive at Gate



Transmit



Regulate Pressure



Distribute



Monitor



Customer Meter

3. Bill & Collect Revenue

Meter Read



Validate



Calculate



Present Bill



Process Payment



Collections



Reconcile

4. Respond to Gas Emergency

Report / Detect



Dispatch



Isolate & Safe



Investigate



Repair



Restore Supply



Incident Review

5. Maintain Pipeline Integrity

Risk Assessment



Inspection Plan



Inspect / Pig



Analyse



Remediate



Verify



Record & Report

Organization Structure

Governing Body

Board of Directors / Regulatory Authority

Executive Leadership

CEO / Managing Director

C-Suite





Business Process Architecture

1. Gas Supply & Trading

1.1 Gas Supply Procurement & Contracting

1.2 Supply Nomination & Scheduling

1.3 LNG / LPG Storage & Peaking

1.4 Gas Quality Monitoring & Control

2. Transmission Operations

2.1 High-Pressure Pipeline Operations

2.2 Compressor Station Management

2.3 Gas Flow Control & Balancing

2.4 Interconnect & Shipper Services

3. Distribution Operations

3.1 Distribution Network Management

3.2 Service Connection & Reinstatement

3.3 Emergency Gas Escape Response

3.4 Odorization & Gas Quality Distribution

4. Asset & Integrity Management

4.1 Pipeline Integrity Assessment (PIMS)

4.2 Cathodic Protection Management

4.3 Mains Replacement Program

4.4 Inline Inspection & Pigging

5. Customer Management

5.1 Service Connection Management

5.2 Meter Reading & Validation

5.3 Billing & Revenue Collection

5.4 Customer Enquiry & Gas Safety Advice

6. Safety & Emergency

6.1 Emergency Response Procedures

6.2 Hazardous Area & Risk Assessment

6.3 Permit to Work & Safety Controls

6.4 Major Accident Hazard Management

Core Data Domains & Key Entities

Customer Domain

Customer

Account

Premise / Property

Service Point

Service Agreement

Tariff Classification

Asset Domain

Pipeline Segment

Compressor Station

Gate / Regulator Station

LNG / Storage Facility

Valve / Fitting

Meter (Residential / Industrial)

Gas Quality Domain

Gas Composition Sample

Calorific Value (CV) Reading

Odorization Level

Pressure / Temperature Reading

Gas Quality Compliance Record

Integrity & Safety Domain

Integrity Assessment Record

Corrosion Inspection Data

Pipeline Risk Assessment

Gas Escape / Incident Report

Permit to Work Record

Operations Domain

Work Order

Service Request

Gas Interruption Event

Emergency Callout Record

Maintenance Record

Metering & Consumption Domain

Meter Reading

Consumption Profile

Smart Meter Data (HHD/AMI)

Demand Forecast

Unaccounted For Gas (UFG)

Financial Domain

Bill

Payment

Invoice

Revenue

Budget / RAB Entry

Regulatory Domain

Gas Safety License

Regulatory Filing

Compliance Report

Tariff & Rate Schedule

Audit Record

Geospatial Domain

Pipeline Network Layer (GIS)

Pressure Tier / Zone

Right-of-Way (ROW)

High Consequence Area (HCA)

Service Territory Boundary

Application Landscape

Customer Systems

Customer Information System (CIS)

Customer Portal (Web & Mobile)

CRM Platform

IVR & Contact Centre System

Gas Safety Notification System

Metering & Billing

Smart Metering System (AMI / SMETS2)

Meter Data Management (MDM)

Billing Engine

Payment Gateway

Collections Management

Gas Control & SCADA

SCADA / Energy Management System (EMS)

Gas Control System (GCS)

Compressor Management System

Pressure Regulation Control

Gas Quality Analyser Platform

Pipeline Integrity & Safety

Pipeline Integrity Management System (PIMS)

Cathodic Protection Monitoring

Leak Detection System (Acoustic / Satellite)

Inspection & Survey Management

Hazardous Area Register

Asset Management

Enterprise Asset Management (EAM)

Geographic Information System (GIS)

Asset Performance Management (APM)

Work & Field Management System

Capital Project Management

Gas Supply & Trading

Gas Nomination & Scheduling System

Gas Trading & Risk Management (GTRM)

LNG / Storage Inventory Management

Balancing & Settlement System

Load Forecasting System

Enterprise Systems

Enterprise Resource Planning (ERP)

Human Capital Management (HCM)

Financial Management System

Supply Chain & Procurement

Document Management System

Analytics & Intelligence

Data Warehouse / Data Lake

Business Intelligence Platform

Pipeline Digital Twin

AI/ML Predictive Maintenance

Methane Emissions Monitoring

Integration & Governance

Enterprise Service Bus (ESB)

API Management Platform

Master Data Management (MDM)

Data Governance Platform

Technology Stack

Compute Infrastructure

- On-Premise Data Centres
- Cloud Infrastructure (AWS / Azure / GCP)
- Hybrid Cloud Architecture
- Virtual Machine Platform
- Container Orchestration (Kubernetes)

Network & Connectivity

- Wide Area Network (WAN)
- Local Area Network (LAN)
- Software-Defined Networking
- MPLS / Leased Lines
- 4G/5G/LTE for Field Operations
- LoRaWAN / Satellite for Remote Pipelines

OT / SCADA / ICS

- SCADA (Transmission & Distribution)
- Programmable Logic Controllers (PLCs)
- Remote Terminal Units (RTUs)
- Historian / Time-Series Database
- Telemetry & Remote I/O
- DNP3 / Modbus / IEC 60870 Protocols

Data Management

- Relational Databases (Oracle, SQL Server)
- NoSQL Databases (MongoDB, InfluxDB)
- Data Lakes (Hadoop, Azure Data Lake)
- Data Warehouses (Snowflake, Synapse)
- Real-time Streaming (Kafka, MQTT)

Security

- Identity & Access Management (IAM)
- Multi-Factor Authentication (MFA)
- Firewalls & IDS/IPS
- SIEM Platform
- Encryption (Data at Rest & in Transit)
- OT/ICS Cybersecurity (IEC 62443)

Integration & Middleware

- Enterprise Service Bus

- API Gateway
- Message Queues (RabbitMQ, Kafka)
- ETL / ELT Tools
- iPaaS (Integration Platform as a Service)

Field Technology

- Smart Gas Meters (SMETS2 / AMI)
- Portable Gas Detectors (LEL / CO)
- Inline Inspection Tools (Smart Pigs)
- Field Mobile Devices & Tablets
- GPS & GIS Mapping Tools
- Satellite Methane Detection (UAV / Orbital)

Disaster Recovery & Backup

- Backup & Recovery Solutions
- Disaster Recovery Site
- High Availability Clustering
- Business Continuity Systems
- Gas Control Room Redundancy

CROSS-CUTTING CONCERNS Governance, Security & Compliance

Security & Privacy

Cybersecurity, OT/ICS Security (IEC 62443), Data Protection

Gas Safety

Emergency Procedures, MAH Management, Permit to Work, Safety Case

Governance

Architecture Board, Standards, Policies

Regulatory Compliance

Gas Safety Regulations, Pipeline Codes (ASME B31.8), Environmental Licences

Data Quality & MDM

Master Data, Governance Rules, Quality Assurance

Change Management

Release Management, Configuration Control, Management of Change (MOC)

Architecture Framework Legend

Strategy Layer: Vision, goals, and strategic objectives

Business Layer: Capabilities, processes, and organization

Information Layer: Data domains, entities, and flows

Application Layer: Systems and applications portfolio

Technology Layer: Infrastructure, platforms, and tools

Cross-Cutting: Security, governance, and compliance

Framework Alignment: This Enterprise Architecture is aligned with TOGAF (The Open

Group Architecture Framework) and follows industry best practices for gas utility

operations. It provides a comprehensive view of the organization's strategy, business,

information, applications, and technology in an integrated manner — covering gas supply,

transmission, distribution, pipeline integrity, safety management, customer operations,

and energy transition programs.